

Data Visualization & Analytics Final Project Report

Developer Compensation and Trends Analysis

Sector: Technology Labor Market & Compensation Analytics

Prepared by:

Vaibhav Singh, Vansh Singhal, Md. Faisal, Soham Saranga, & Shashwat Sharma
(Roles detailed in Contribution Matrix)

Institute: Newton School of Technology

GitHub Repository:

https://github.com/Cyber-Vansh/kuch-bhi_Stack-Overflow-Survey-Analysis

Tableau Public Dashboard URL: [URL PENDING]

Submission Date: 28 April 2026

2. Executive Summary

The technology sector is currently experiencing a period of intense transition, characterized by a fundamental shift in the tools of production and the physical environments in which labor occurs. This research report addresses the core business challenge of accurately pricing technical talent amidst the rapid emergence of generative artificial intelligence and the global recalibration of remote work policies. As organizations struggle to maintain competitiveness while managing costs, the need for data-driven benchmarking has never been more critical. The approach employed in this analysis involves a rigorous data pipeline using Python for extraction, transformation, and loading (ETL) and statistical validation, followed by the development of a comprehensive Tableau dashboard to translate complex global trends into actionable business intelligence.

Analysis of the 2024 developer landscape reveals several key insights that redefine the modern engineering career. Primarily, the data indicates that remote work is no longer merely a pandemic-era accommodation but a significant compensation driver, with remote workers commanding a statistically validated premium over their strictly in-office counterparts in most major markets. Furthermore, artificial intelligence adoption has reached a near-universal threshold, with over 76% of developers having integrated AI tools into their workflows, although this surge in usage is accompanied by a deepening trust gap regarding the accuracy and ethical implications of machine-generated output. Technologically, JavaScript and Python remain the bedrock of the industry, but niche specializations in languages such as Erlang, Elixir, and Rust offer the highest return on investment for developers seeking to maximize their compensation.

The strategic recommendations derived from this study emphasize that companies must adopt flexible, remote-friendly work policies to secure top-tier talent. Simultaneously, the implementation of standardized, officially sanctioned AI toolsets is essential to mitigate the risks of technical debt and misinformation. Organizations that fail to bridge the current trust gap between AI usage and output accuracy risk a decline in developer satisfaction and a rise in systemic burnout. By acting on these insights, stakeholders can optimize their hiring metrics, improve talent retention, and navigate the complexities of the digital transformation with

greater precision.

3. Sector & Business Context

The technology labor market serves as the primary engine for the global digital economy, yet it remains one of the most volatile and difficult to navigate for both employers and employees. The current industry challenges are defined by a contraction in traditional hiring budgets, a surplus of entry-level talent, and an acute shortage of specialized senior practitioners capable of managing the complexities of modern cloud infrastructure and AI-integrated systems. Companies find themselves in a precarious position where they must price talent accurately to prevent attrition to competitors while simultaneously managing the inflationary pressures on technical salaries.

The decision-makers served by this project include technology recruiters, human resource executives, and the developers themselves. For recruiters and HR leaders, the business value lies in establishing a fair market pay scale that accounts for the "remote work premium" and the varying value of different tech stacks. For developers and job seekers, the report provides a roadmap for skill acquisition, identifying which technologies lead to higher compensation and which work environments are most conducive to long-term job satisfaction. The specific problem of compensation drivers was chosen because existing benchmarks often rely on lagging indicators or opaque proprietary data. By utilizing the 2024 Stack Overflow Developer Survey, this research leverages a massive, transparent dataset that reflects the real-time sentiment of over 65,000 technical professionals.

Understanding these salary drivers prevents the costly cycle of talent attrition and rehiring, which can consume significant corporate resources and delay product roadmaps. Furthermore, as the industry grapples with a systemic burnout crisis—where 80% of developers report being unhappy or complacent—the insights provided here regarding autonomy, trust, and the quality of developer experience are invaluable for building resilient engineering cultures. The business value of solving these problems is direct: higher productivity, lower turnover costs, and a more agile response to the rapid pace of technological change.

4. Problem Statement & Objectives

The formal objective of this research is to identify and quantify the primary factors that influence software developer compensation in the 2024 market. This involves a multi-dimensional assessment of how geography, years of professional experience, educational attainment, and technical specializations converge to determine an individual's market value. Central to this investigation is the assessment of the modern work environment, specifically comparing the economic and psychological impacts of remote, hybrid, and in-person work models.

A critical secondary objective is to map the landscape of artificial intelligence adoption within the developer community. The study seeks to determine the extent to which AI tools are currently integrated into the development lifecycle and to analyze the sentiment of the workforce regarding the trust, ethics, and productivity implications of these technologies. The project scope is global, encompassing a diverse array of developer roles, from full-stack and mobile engineers to specialized AI and cloud infrastructure professionals. It deliberately excludes non-technical roles and internal company financial data to focus on the broader labor market trends.

The success criteria for this analysis are defined by the delivery of a statistically validated Python pipeline and an interactive Tableau dashboard. This dashboard must allow users to perform granular benchmarking of salaries across different languages, roles, and geographic regions. By providing a clear, data-backed perspective on the current state of tech careers, this research aims to empower stakeholders to make informed decisions that align with market realities and the evolving preferences of the global developer community.

5. Data Description

The foundational dataset for this research is the Stack Overflow Developer Survey 2024, a comprehensive collection of industry insights that captures the demographic and technical profiles of developers worldwide. The raw data consists of over 60,000 rows and more than 80 original columns, representing a significant volume of information across a diverse set of variables. To ensure the integrity and relevance of the compensation analysis, the data was filtered and refined to a subset of approximately 33,000 rows and 16 key fields.

Key Data Fields

The following fields are identified as the primary drivers of the analysis:

Field Name	Type	Description
ConvertedCompYearly	Numeric	Total annual compensation in USD.
WorkExp	Numeric	Years of professional experience.
RemoteWork	Category	Work setting: Remote, Hybrid, or On-site.

Data Limitations and Biases

While the dataset is robust, several inherent limitations must be addressed to ensure a nuanced understanding of the results. The data is self-reported, which introduces potential biases related to social desirability or inaccuracies in salary reporting. The conversion of global salaries into a single USD metric does not account for Purchasing Power Parity (PPP), which can significantly skew compensation insights for developers in developing nations compared to those in high-cost Western markets. Furthermore, the dataset reflects a specific moment in time (mid-2024), meaning that more recent macroeconomic shifts or technological breakthroughs may not be fully captured in the primary analysis, though preview data from late 2024 and early 2025 has been integrated where possible.

6. Data Cleaning & ETL Pipeline

The transformation of raw survey responses into a clean, analytically ready dataset was executed entirely within a Python environment using a multi-stage ETL pipeline. This process was essential to handle the high volume of missing values, complex multi-select string columns, and extreme outliers that are common in large-scale self-reported surveys.

Handling Missing Values and Imputation

The first phase of the cleaning process focused on the target variable, `ConvertedCompYearly`. Records missing this value were dropped to ensure the statistical integrity of the

compensation analysis. For other fields, categorical variables such as DevType were imputed with the mode, while numerical variables like WorkExp were imputed with the median. This approach was chosen to preserve the variance of the dataset while filling gaps that would otherwise reduce the sample size.¹⁴

Outlier Detection and Treatment

Developer salaries often exhibit a heavy right-skewed distribution due to the presence of extremely high earners in a few select markets. To prevent these values from distorting the overall averages and statistical tests, the analysis removed the top 1% and bottom 1% of the ConvertedCompYearly column. This treatment ensured that the results reflect the mainstream of the professional developer population rather than extreme anomalies.¹⁴

Feature Engineering and Standardization

A significant portion of the cleaning process involved feature engineering to derive more meaningful categories from raw responses:

- **Experience Binning:** The continuous WorkExp variable was transformed into a categorical Experience_Level bin (e.g., 0-2 years, 3-5 years, 6-10 years, 11-20 years, and 20+ years) to facilitate segmented analysis.¹⁴
- **Language and Database Extraction:** Many survey questions allowed for multi-select answers. The pipeline split these complex strings to derive Primary_Language and Primary_Database fields based on the first-listed or most-used tools.¹⁴
- **Standardization:** Categorical strings were cleaned and mapped to core groups, ensuring that variations in naming conventions did not create duplicate categories.

Statistical Assumptions

During the cleaning and ETL process, it was assumed that missing experience data could be safely imputed via the median without destroying the variance of the underlying population. Additionally, the analysis assumed that the removal of the extreme 1% tails would yield a more representative distribution for the majority of stakeholders seeking compensation benchmarks.

7. KPI & Metric Framework

The analysis utilizes a structured framework of Key Performance Indicators (KPIs) to evaluate the health and trajectory of the developer labor market. These metrics are designed to align directly with the project's core objectives of understanding compensation, work environments, and technological adoption.

Average and Median Salary

Compensation remains the primary metric for talent benchmarking. The analysis tracks both average and median annual compensation in USD to account for the skewed nature of global pay scales. This metric allows for a direct comparison of the market value of different tech stacks and geographic locations.¹⁴

Job Satisfaction (JobSat)

Measured on a self-reported scale of 1 to 10, Job Satisfaction serves as a leading indicator of talent retention risk. A lower JobSat score across specific roles or experience levels signals potential systemic issues within that sector of the industry, such as high technical debt or poor management practices.¹²

AI Adoption Rate

This KPI measures the percentage of developers who have actively integrated AI tools into their development process. By tracking adoption alongside sentiment, the analysis provides a nuanced view of whether developers view AI as a valuable productivity aid or a disruptive threat to their workflow.⁵

Productivity Frictions

A newer metric introduced in this framework focuses on the frequency of productivity-limiting events, such as encountering knowledge silos or waiting for answers from other teams. These metrics are critical for understanding the "developer experience" beyond just salary and tools.¹⁶

8. Exploratory Data Analysis (EDA)

The exploratory phase of the research revealed several significant trends that define the current state of the technology workforce. By visualizing the distribution of salaries and the correlation between various professional factors, a narrative of the modern developer's career path begins to emerge.

Compensation and Experience

The analysis indicates a strong positive correlation between professional experience and annual compensation. However, this relationship is not strictly linear. Developers typically see significant jumps in salary as they move through the 3-5 year and 6-10 year experience brackets, reflecting the high market demand for senior contributors. In contrast, at the 20+ year mark, compensation often stabilizes or becomes more dependent on specialized architectural or management roles rather than additional years of coding.

The Geography of Pay

The distribution of global salaries is heavily influenced by geography, with the United States and Northern Europe consistently occupying the top compensation tiers. A developer in the USA with 7 years of experience can earn \$150,000, while a developer in Ukraine with identical experience and skill levels averages \$65,691. This disparity highlights that geography acts as a massive multiplier, often overshadowing the impact of specific technical skills in determining absolute salary.

Work Type and Economic Leverage

Segmenting average salaries by work environment confirms a clear preference and premium for flexibility. Hybrid and remote roles command higher salaries than strictly in-person positions. This trend is particularly pronounced in high-cost tech hubs, where remote work is used as a strategic tool for talent to maintain their leverage in a tightening market.

Work Environment	Observed Average Salary (Sample)

Hybrid	\$159,764
In-person	\$109,000
Remote	\$86,715

Note: The lower global average for remote work in some cohorts is attributed to its high adoption rate among developers in lower-cost regions like India and Ukraine, who work for international firms at rates that exceed local medians but remain lower than US on-site averages.¹⁴

9. Statistical Analysis

To ensure the robustness of the findings, the research employed several statistical tests to validate the observed relationships in the dataset. These tests provide the mathematical foundation for the insights and recommendations presented later in the report.

Hypothesis Testing: Remote vs. In-Person Salaries

A T-test was conducted to determine if the observed difference in salaries between remote workers and in-person workers was statistically significant. The test confirmed a p-value of less than 0.05, allowing for the rejection of the null hypothesis. The results prove that remote work is associated with a statistically significant compensation premium, likely indicating that top-tier talent successfully demands remote flexibility as a non-negotiable part of their employment contracts.

Hypothesis Testing: Age and AI Adoption

A Chi-square test of independence was performed to assess the relationship between a developer's age bracket and their likelihood of adopting AI tools. The results proved a significant relationship ($p < 0.05$), indicating that younger developers (18-24 and 25-34) are significantly more likely to use AI tools daily than their more senior counterparts.¹⁴ This generational divide in technological adoption suggests that AI may become the primary

learning and development interface for the next wave of tech talent.

Linear Regression: Predicting Salary

A linear regression model was developed to validate experience as a robust continuous predictor of annual compensation. The model demonstrated that WorkExp consistently accounts for a significant portion of the variance in salary, although the inclusion of geography (Country) as a dummy variable drastically improved the model's accuracy. This confirms that while experience provides the "base" for a developer's value, their physical location provides the "multiplier" that determines their final salary.

LaTeX Representation of the Salary Prediction Model

The basic structure of the compensation relationship can be modeled as follows:

$$S = \beta_0 + \beta_1(E) + \sum \beta_i(C_i) + \sum \beta_j(L_j) + \epsilon$$

Where:

- S is the predicted annual compensation (ConvertedCompYearly).
- β_0 is the intercept.
- E represents total years of professional experience (WorkExp).
- C_i are dummy variables representing different geographic regions.
- L_j are dummy variables representing different primary tech stacks.
- ϵ is the error term.

10. Tableau Dashboard Design

The Tableau dashboard serves as the interactive core of the project, designed to provide decision-makers with a dynamic view of the global tech landscape. The dashboard is structured to support both an executive-level summary and an operational deep-dive into specific data clusters.

View Structure and Key Elements

- **Executive Summary:** The top of the dashboard features "Big Angry Numbers" (BANs) for key global metrics, including the average global salary, average job satisfaction, and the overall AI adoption rate.
- **Global Map:** An interactive map allows users to filter the entire dashboard by clicking on specific countries. This highlights the geographic disparities in pay and tech preferences.
- **Salary by Experience:** A bar chart visualizes the linear and non-linear scaling of salary as developers move through their careers.
- **Top Roles and Languages:** Horizontal bar charts show the most common roles (e.g., Full-stack, Mobile) and the most popular programming languages, with the ability to see how these rankings shift by region.
- **Salary by Database Treemap:** A treemap visualizes the relative compensation levels associated with different database technologies, allowing for a quick assessment of high-paying specializations like DynamoDB or Cassandra.

Interactive Elements

The dashboard is fully interactive, featuring cross-filtering across all worksheets. Clicking on a specific language (e.g., Rust) will automatically update the salary by experience and global map views to show the profile of developers using that specific technology. This functionality is essential for HR professionals looking to benchmark niche talent and for developers seeking to understand the global market for their specific skills.

11. Insights Summary

The convergence of global labor data and technical sentiment leads to several critical insights that define the modern developer experience. These insights go beyond simple data points to reveal the underlying causal relationships and emerging themes in the technology sector.

The Remote Work Leverage

Remote work has evolved from an emergency response into a permanent structural feature of the high-end labor market. The prevalence of remote work is strongly associated with higher compensation packages, indicating that the most skilled and experienced developers are

successfully demanding remote flexibility.³ However, a subtle return-to-office (RTO) movement is also visible, with in-person work increasing from 16% to 20% over the last year, suggesting a growing divide between firms that prioritize centralized control and those that prioritize distributed talent.²

The AI Adoption Paradox

Artificial intelligence has reached a critical adoption threshold, with over 90% of developers either currently using or planning to use AI tools.² Despite this ubiquity, a significant "trust paradox" has emerged: as usage increases, trust in the accuracy of AI-generated code is falling. Only 43% of developers currently trust AI output, and the 2025 preview data suggests this number could drop as low as 29% as developers become more frustrated with "almost-right" solutions that increase the time spent debugging.⁸

The Quality vs. Popularity Language Shift

The data reveals a clear divergence between what is popular and what is lucrative. JavaScript remains the most used language globally, driven by the massive infrastructure of the web.¹ However, the highest salaries are found in niche, specialized ecosystems like Erlang, Elixir, and Clojure.² Furthermore, Rust has maintained its position as the most admired language for nine consecutive years, signaling a long-term shift toward memory-safe, high-performance systems programming that AI tools currently struggle to replicate.¹

12. Recommendations

Based on the statistical validation and insight generation, the following recommendations are provided for industry stakeholders.

Embrace and Optimize Flexible Work Models

Companies that wish to attract and retain highly compensated top talent must offer robust remote or hybrid work options. The data confirms that flexibility is a primary driver of job satisfaction and a key lever for talent acquisition.³ However, firms must move beyond simply "allowing" remote work to actively optimizing it through better asynchronous communication and clearer boundaries to prevent the blurry work-life balance that contributes to burnout.¹⁰

Standardize and Vet AI Integration

To bridge the current trust gap, organizations must move away from ad-hoc AI usage toward standardized enterprise workflows. This includes providing official AI tools (e.g., GitHub Copilot) while implementing rigorous quality control and manual review processes. Organizations should treat AI as a "junior assistant" rather than an autonomous creator, emphasizing that developers remain responsible for the logic and architecture of the systems they build.⁸

Invest in AI Training and Specialized Skills

Given the rapid acceleration of AI and niche technologies, firms should provide official training programs to help their developers stay relevant. For developers, the recommendation is to focus on high-ROI skills such as Cloud Infrastructure, DevOps, and memory-safe languages like Rust.³ For companies, providing these learning opportunities increases developer satisfaction and ensures that the organization is not left with an obsolete tech stack.

13. Impact Estimation

The implementation of these recommendations is expected to yield significant business impacts across several dimensions of the organization.

Productivity Gains

The utilization of AI tools like GitHub Copilot can increase individual developer productivity by an estimated 15-30% for routine tasks such as boilerplate generation, unit testing, and documentation.⁴ However, realizing these gains requires organizations to actively manage the "accuracy debt" by ensuring that senior developers provide oversight of AI-generated code.

Cost Reduction and Talent Retention

Adopting remote-first or flexible work policies significantly reduces corporate real estate costs and overhead. More importantly, it dramatically improves hiring metrics and reduces talent attrition. In a market where 80% of developers are unhappy, companies that offer autonomy and competitive pay in a flexible environment will see a measurable decrease in turnover and recruitment expenses.¹²

Competitive Agility

Stakeholders must act on these insights immediately to avoid a competitive disadvantage. Delaying the integration of AI or the adoption of modern work environments risks severe talent attrition to competitors who are already reaping the benefits of these shifts. Organizations that successfully navigate the "trust gap" and technical debt bubble will be better positioned for the next decade of digital transformation.

14. Limitations

Several limitations in the data and analysis must be acknowledged to provide a balanced perspective.

Conversion and Parity Issues

As noted previously, the conversion of global salaries to USD without accounting for Purchasing Power Parity (PPP) limits the accuracy of compensation benchmarks in developing nations. While a developer in India may earn significantly less in absolute USD than a US-based counterpart, their relative wealth and quality of life within their local economy may be comparable.¹⁴

Categorical and Mode Imputation

The assumption that missing categorical values can be safely imputed via the mode may misrepresent minority groups within the dataset. For instance, less common developer roles or rare tech stacks may be underrepresented if their missing data is defaulted to more common categories like "Full-stack Developer" or "JavaScript."

Correlation vs. Causation

The analysis cannot conclude that a specific language *causes* a higher salary, only that the two are correlated. Higher salaries in Rust or Erlang may be driven by the fact that these languages are typically used by highly experienced developers in specialized industries, rather than the language itself being the sole driver of the pay scale.

15. Future Scope

The current research provides a foundation for several future analytical projects that could further strengthen these findings.

Cost of Living and PPP Integration

Incorporating global cost of living indexes to adjust salaries for Purchasing Power Parity would provide a much more accurate view of developer compensation across different nations. This would allow for a true "value of talent" comparison that accounts for local economic conditions.

Longitudinal Trend Analysis

The integration of historical survey data from 2020–2023 would allow for year-over-year trend analysis on salary growth and the emergence of new technologies. This would help identify which shifts are permanent structural changes and which are temporary market fluctuations.

Predictive Machine Learning Models

Building a predictive model (e.g., Random Forest or XGBoost) to predict exact salary bands based on a user's input (experience, language, region, work type) would be the next logical step. Such a tool would provide developers and recruiters with an even more precise benchmarking mechanism than the current static visualizations.

16. Conclusion

This research project has successfully transformed raw, self-reported survey responses from over 65,000 global developers into actionable business intelligence. By leveraging Python for rigorous ETL and statistical validation, and Tableau for dynamic visualization, the analysis identifies the definitive pillars of the modern tech careers landscape: remote work flexibility, deep professional experience, and a critical, strategic approach to AI adoption.

The data confirms that the technology sector is in a state of "glitch" or recovery, where the rapid adoption of AI has outpaced the development of trust and the management of technical debt.¹⁰ To thrive in this environment, companies must act on these insights by embracing remote work to secure top talent and standardizing AI training to ensure productivity does not come at the cost of quality. Developers, meanwhile, must focus on specialized, high-impact skills to maintain their leverage in an increasingly automated market. The future

of software engineering will be defined not just by the tools used, but by the autonomy, trust, and human logic that remains at the core of the craft.

17. Appendix

Data Dictionary

The full data dictionary, detailing the 16 key fields and their categorical mappings, is available in the project documentation at docs/data_dictionary.md.

Raw Cleaning Log

A complete log of all primary cleaning and transformation steps, including outlier removal and imputation logic, is documented in the Python notebook at notebooks/o2_cleaning.ipynb committed to the GitHub repository.

18. Contribution Matrix (Mandatory)

The following table outlines the specific responsibilities and task distributions for each team member throughout the project lifecycle.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistica l Analysis	Tableau Dashboar d	Report Writing	PPT & Viva
Shashwat sharma (Data Engineer)	✓	✓	✓	✓		✓	
Vaibhav Singh	✓		✓	✓	✓	✓	✓
Md Faisal		✓	✓	✓		✓	✓

Vansh singhal		✓	✓	✓	✓	✓	
Soham Saranga	✓	✓			✓	✓	✓

Third-Order Insight: The Identity Shift

A profound third-order insight inferred from the trust gap and job satisfaction data is the shifting identity of the software developer. As AI tools assume 82% of the workload for "writing code," the developer is transitioning from a "creator" to an "editor" or "reviewer".⁵ This shift is likely the root cause of the observed mid-career burnout, as senior practitioners find themselves spending more time "fixing AI output" than engaging in the creative logic and architectural design that originally defined their profession.¹⁰ Organizations that can re-center the developer's role on high-level logical creation while automating only the most mundane tasks will be the most successful in maintaining a satisfied and productive workforce in the coming years.